

BNY MELLON – INTERNSHIP INTERVIEW EXPERIENCE

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About the company

BNY is a global financial services company headquartered in New York City. BNY oversees nearly \$50 trillion in assets for our clients – managing it, moving it, and keeping it safe. For 240 years, they have been creating new solutions that benefit businesses, communities, and people everywhere. It is the world's largest custodian bank and securities services company. The bank's primary functions are managing and servicing the investments of institutions and high-net-worth individuals. Its two primary businesses are Investment Services and Investment Management. The bank's clients include 80 percent of Fortune 500 companies. Currently, the company has over 12,000 employees across our three locations: Mumbai, Pune, and Chennai

No of Rounds: 3 (Online Test + Technical interview + Technical & HR interview)

Stipend: 75,000 INR per month

Duration of internship: 8-10 weeks

SCHEDULE:

- **ONLINE ASSESSMENT (16/08/24)**

Coding Round: Conducted in Hacker-rank Website

Test Duration: 120 mins Camera On 1 easy, 2 medium and 1 hard question.

1. BALANCING ELEMENTS

When an element is deleted from an array, the higher-indexed elements shift down one index to fill the gap, A “balancing element” is defined as an element that, when deleted from the array, results in the sum of the even-indexed elements being equal to the sum of the odd-indexed elements. Determine how many balancing elements a given array contains.

Example: Input: $n=5$ arr = [5,5,2,8]

Output: 2 When the first or second 5 is deleted, the array becomes [5, 2, 5, 8].

The sum[even] = $5+5=10$ and sum[odd] = $-2+8=10$.

No other elements of the original array have that property.

There are 2 balancing elements: arr[0] and arr[1].

2. MINIMUM FUEL COST

There are n cars located on a 2 dimensional plane at positions $(x[i], y[i])$ where $0 \leq i \leq n$.

They need to be parked in a straight line parallel to the x axis with no spaces between them. The fuel consumed to move a car is $\text{abs}(x[\text{finish}] - x[\text{start}]) + \text{abs}(y[\text{finish}] - y[\text{start}])$.

Determine the minimum fuel cost to arrange the cars side by side in a row parallel to the X axis

Example: $X=[1,4]$, $Y=[1,4]$ Total fuel : $2+3=5$

3.PRODUCT DEPENDENCY TREE

Given a network of product dependencies represented as a tree. The tree consists of `treeNodes` vertices and the relationships between products are defined by two arrays: `treeFrom` and `treeTo`, both of size `treeNodes - 1`. These arrays specify the connections, where `treeFrom[i]` is connected to `treeTo[i]`.

For this product dependency tree, the task is to determine the number of triplets of vertices (i, j, k) that meet the following conditions :

$0 \leq i < j < k < \text{treeNodes}$

There should be no simple path connecting vertices i, j and k .

Due to the potentially large number of triplets, return the answer module $(10^9 + 7)$

4.PORTFOLIO BACKTESTING :

The traders of Hacker-land have stock prediction model that they want to backtest. The predicted profit earned every month by a certain portfolio is given by an array `pnl` of n integers where a negative integer denotes loss.

The traders decide to iterate through the months from 1 to n , and if the `pnl` is positive, they add it to their profit and if the `pnl` is negative, they can either subtract it or skip it. The net profit must always be greater than or equal to 0.

Given the array `pnl` and an integer k , find the maximum total profit that can be earned by the portfolio after subtracting at least k losses.

If it is not possible to take at least k losses, report -1 as the answer.

Example: Suppose $n=5$, $k=2$ and `pnl`=[4, -3, 6, -2, -1]

The optimal strategy is to add the profits 4 and 6, and the losses [-2, -1] to get a net profit of 7 which is the maximum possible.

All the testcases were passed for the first 2 questions

10/15 for the 3rd question

2/15 for the 4th question

24 Students (CEG and MIT) were selected after the online coding test.

The interview was held at CUIC on 17/08/2024

- **PPT (PRE-PLACEMENT TALK) - (9.30 – 10.30)**

The Company People explained us about the work culture and history behind BNY. They told us when was the company started and talked about the founder.

How will the intern experience be in BNY? How well you can groom yourself by doing an internship there? Also, the director briefly explained about the company's ethics and how he started his journey in BNY. Since it is a FINTECH company, He asked whether any of us are familiar with stock marketing. Once the ppt session was over, we were told to wait for 10-15 minutes and we were called for the technical round 1.

- **TECHNICAL INTERVIEW – (10.45 – 11.30)**

There were nearly 5-6 interviewers and 24 students were split up and sent to each room. Fortunately and unfortunately, I was called the first and I went into the room with my resume. The interviewer greeted me and started to see my resume.

He told me to give an introduction about myself.

I explained my journey from 10th std. I added information about my schooling and why I chose computer science branch.

He further asked "Why not IIT and why did u choose CEG? He asked me if I have cleared JEE MAINS". Moving on to the technical Experience, I explained about the DBMS project which we have developed during our 4th semester as a part of our curriculum. I explained about the tables we have used and drew the ER diagram and explained about all the functionalities (triggers, procedures) we have added in this project.

Coding questions :

- a) Write the code snippet for finding the Fibonacci series in all different approaches (Iterative and recursive)
- b) What are linked list? What are its advantages ?
- c) Write a code snippet to delete a specific node in a singly linked list .
- d) To count the number of nodes in a singly linked list.

Write all the approaches and explain the code line by line.

He asked me some new unexpected questions.

- a) What is DNS ?
- b) Explain me completely how a webpage works?
- c) Explain the complete process of getting into a particular website from the home page of chrome . As he was interested in this section ,he further added new questions.
- d) He asked me to explain about cookies and how it works
Since I was not completely dealt with this topic, I was unable to answer some of the questions and admitted to him that I will read on this topic in the future .

That's perfectly fine if you are not able to answer a question, feel free to admit that and move on to the new question.

Constantly address your strengths to the interviewer. The interview can totally turn out to be q/a session on what you mention to him/her.

I further started talking about the clubs and organisations I have been part of.

As I am part of NSS, I mentioned him about my orphanage experience at (SEVALAYA).

It was a coincidence and he informed me that BNY also offers funds to the same orphanage

and he was interested in asking about my personal experience in the orphanage and social camps.

Moving on he asked me "If I had any dream other than working in a big company by clearing all interviews and asked me to explain what the dream is and how am I working to reach that specific goal."

Finally Asked whether I had any questions to ask him

I asked him "How did you feel when you were sitting just like me now during his interview and he shared all his experience."

The interview ended.

Out of 24 students, 10 of them were selected for the next round which was a mix of technical and HR round.

- **TECHNICAL INTERVIEW + HR INTERVIEW (1.00 – 1.40)**

The interviewer asked me to give my introduction and also asked me to explain the projects that I have mentioned in the resume.

He asked me to explain the DBMS project and asked me if I had deployed my project or not.

Questions about the Project:

- a) What is the difficulty you faced while developing the project? How did you overcome it?
- b) What extra special features you have added in your project?
- c) Is the project have a real - life application and where is it currently being used?
- d) Explain the functionalities added in the project.
- e) How much time did you take to develop the project?

Coding Question :

Write the full code along with the driver function to detect a loop in linked list

The code should pass all the edge cases .

Once I wrote the code ,he asked me to explain the code line by line .

<https://leetcode.com/problems/linked-list-cycle/>

Puzzle Question :

<https://www.geeksforgeeks.org/puzzle-7-3-bulbs-and-3-switches/>

I came up with a complex solution and he told me to simplify and give an easy solution

Behavioural Questions:

- a) What did you do to prove yourself when everyone was not believing in you?
- b) If you are a head of 2 organisations, how would you manage yourself and complete the work in both of the sectors and how would you mentally prepare yourself?
- c) What is the one thing that always pushes you to move forward in life?
- d) Can you describe an incident where you were faced with a challenge that was beyond your current abilities? How did you approach the situation, and what was the outcome?
- e) One good reason to hire you?

Finally, the interviewer asked me, do you have any questions? I asked about his journey in BNY .

The interview ended.

And 6 students were offered an internship

PREPARATION TIPS:

I chose C++ for solving problems.

DSA : <https://takeuforward.org/strivers-a2z-dsa-course/strivers-a2z-dsa-course-sheet-2/>

OS : <https://www.geeksforgeeks.org/oops-interview-questions/>

DBMS : <https://www.geeksforgeeks.org/commonly-asked-dbms-interview-questions/>

OOPS : <https://www.geeksforgeeks.org/oops-interview-questions/>

Be Confident on what you put in your resume. Try to understand the question and come up with your solution. If u do not arrive at a solution , feel free to admit to the interviewer and move on to the next question. Try to score up in the next upcoming question . Inform the interviewer about your strong subjects so that he would focus on questioning in that particular subject . Be Clear in your points and try to explain it with a pen and paper in a diagrammatic way.

Thanks !.

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